

Multiple Choice (10 marks)

1. The body mass index (BMI) of vegetarians differs on average from meat-eaters by how much?
 - (a) 3 units of BMI
 - (b) 0.5 units of BMI
 - (c) 1 unit of BMI
 - (d) 2 units of BMI
2. Which of the following statements about glucose metabolism is correct?
 - (a) Glucagon increases the rate of glycolysis
 - (b) Glycolysis requires NADP+
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) The end-product of glycolysis in red blood cells is pyruvate
3. Which vitamin is required for synthesis of the blood clotting proteins?
 - (a) Vitamin A
 - (b) Vitamin D
 - (c) Vitamin E
 - (d) Vitamin K
4. Observational epidemiology studies have produced useful in identifying health effects of many food contaminants, but not for assessing risk, because:
 - (a) It is never possible to know whether observed effects are caused by the contaminant
 - (b) Human exposure through food is usually much less than exposures that have been studied
 - (c) Controlled animal studies provide more reliable information
 - (d) Reliable dose-response data are difficult to acquire and often unavailable
5. Which of the following will result in an increase in the rate of tissue protein synthesis?
 - (a) Increased glucagon secretion in the fasting state.
 - (b) Increased insulin secretion in the fed state.
 - (c) Increased availability of lysine in the fed state.
 - (d) Secretion of cortisol under conditions of stress.

True / False (10 marks)

Answer True or False

6. True or False: The complete oxidation of glucose to carbon dioxide and water yields 25 molecules of ATP.
7. True or False: Low-carbohydrate diets were previously not recommended because they led to less effective weight loss compared to other diets.
8. True or False: The perception of risk significantly influenced consumer choices regarding meat after the outbreak of mad cow disease.
9. True or False: LCD provides less energy than VLCD and usually includes natural foods.
10. True or False: A sedentary 43-year-old woman who weighs 63 kg needs approximately 1350 kcal per day to maintain her energy balance.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between obesity and the risk of endometrial cancer, focusing on the role of hormones such as oestrogen in this process.
12. Discuss the differences between type 1 and type 2 diabetes, focusing on the role of autoimmune factors in their pathogenesis and the implications for treatment and management.

End of Paper